MCS 320 Quiz 8 Friday 23 October 2026

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. Let $f(x, y, z) = xz^2 - y^3 + 3yz + 1 = 0$.
   Compute the points on the surface defined by f closest to the origin.

2. Consider the curve defined by $4\left(x^2 + (y+1)^2 - 1\right)^2 + (y^2 - x^2)(y+1) = 0$.
   (a) Plot the curve for a suitable range of x and y.
   (b) Transform into polar coordinates and plot.
   (c) Describe the differences between the two plots.